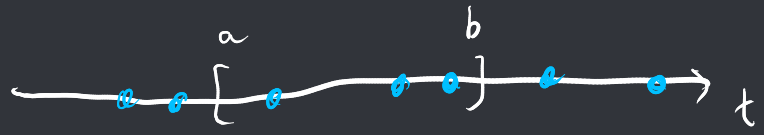
First midterm thursday: Ch. 1 stuff only (discrete time Markov chains)

Today:

- Poisson process continued: conditioning
- Some martingales for Poisson process & random walk, Doob's inequality.
- Problems from Ch. 2.

Conditioning property of Poisson process



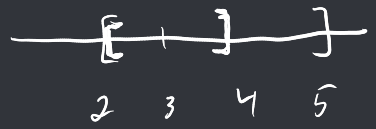
condition on # of points = k
Where are they located?

same distribution as k i.i.d. uniform [a,b] points.

Usage: suppose point process X w/ intensity λ.

Find $(*)\ P(X([2,4]) = 2 \mid \underbrace{X([2,5]) = 3}_{})$.

points in $[2,5]$ have unif $[2,5]$ distn, independent



Each point is in $[2,4]$ indep. w.p. $\frac{4-2}{5-2} = 2/3$

$$(*) = P(\mathrm{Bin}(3, 2/3) = 2)$$

$$= \binom{3}{2}\left(\tfrac{2}{3}\right)^2\left(\tfrac{1}{3}\right).$$

☆ Martingales for Poisson processes

Recall in discrete time, random walk has martingales of the form

$$X_n - n\mu, \qquad X_n^2 - n$$

$\mu = 2p - 1$ $\qquad$ $(p = 1/2)$

- linear
- quadratic
- exponential $\qquad r^{X_n} \qquad r = \left(\frac{1-p}{p}\right)$

Similar martingales exist for Poisson process.

Linear: if $(X_t)_{t \in \mathbb{R}_+}$ is Poisson process w/ intensity λ then

$$M_t := X_t - \lambda t$$

is a martingale because

$$E\left[X_{t+h} - X_t \,\middle|\, (X_s)_{s \le t} \right] = \lambda h$$

so

$$E\left[M_{t+h} \,\middle|\, (X_s)_{s \le t} \right] = M_t + \overbrace{E\left[X_{t+h} - X_t \,\middle|\, (X_s)_{s \le t} \right]}^{= \lambda h} - \lambda h$$

$= M_t.$

<u>Quadratic</u>: what can we add or subtract from X_t^2 in order to obtain a martingale?

"$\Rightarrow$" In continuous time, a martingale is a process with $\underbrace{\text{"average velocity"}}$ $= 0$.

e.g. $\lim_{h \to 0} E\left[\frac{X_{t+h}^2 - X_t^2}{h} \,\middle|\, (X_s)_{s \le t}\right]$ is "average velocity" of X_t^2.

$\uparrow$

for small h, $P(X_{t+h} - X_t \geq 2) = O(h^2)$

so

$$E\left[\frac{X_{t+h}^2 - X_t^2}{h} \,\middle|\, (X_s)_{s \leq t}\right] \approx \overbrace{P(X_{t+h} \neq X_t)}^{\approx \lambda h} \cdot \frac{(2X_t + 1)}{h}$$

$$\approx (2X_t + 1)\lambda$$

<u>Ans</u>: it turns out $(X_t - \lambda t)^2 - \lambda t$

is a martingale (consistent with increment distribution since

$$E\left[(X_t - \lambda t)^2\right]$$
$$= E\left[(X_t - EX_t)^2\right]$$
$$= Var(X_t) = \lambda t$$

Exponential martingales for Poisson process

given $\theta > 0$, want $c(\theta)$ such that

$$e^{\theta X_t - t c(\theta)}$$ is a martingale.

To find $c(\theta)$, note from martingale property

$$E\left[e^{\theta X_t - t c(\theta)}\right] = e^{\theta X_0 - 0 \cdot c(\theta)} = 1.$$

so

$$\underbrace{E\left[e^{\theta X_t}\right]} = e^{t c(\theta)}$$

if $X_t \overset{d}{=} \text{Poiss}(\lambda t)$ then

$$P(X_t = k) = e^{-\lambda t} (\lambda t)^k / k!$$

so

$$E\left[e^{\theta X_t}\right] = \sum_k e^{\theta k} e^{-\lambda t} (\lambda t)^k / k!$$

$$= e^{-\lambda t} \sum_k (\lambda t e^{\theta})^k / k!$$

$$= e^{\lambda t (e^{\theta} - 1)}$$

i.e. take $c(\theta) = \lambda(e^{\theta} - 1)$.

Q: what are these good for?

A: exponential functions lead to strong <u>tail estimates</u> via <u>Markov's inequality</u>.

$P(X > a)$, large a

if $X \geq 0$ then $P(X \geq a) \leq E[X]/a$

$$X \geq a\,\mathbb{1}(X \geq a) \qquad \text{(for } a>0\text{)}$$

(requires $X \geq 0$)

$$\therefore \quad E[X] \geq a\,\underbrace{E[\mathbb{1}(X \geq a)]}_{P(X \geq a)}$$

strengthening Markov's inequality via exponential functions

for $\theta > 0$, $X \geq a \iff e^{\theta X} \geq e^{\theta a}$ so

$$P(X \geq a) = P(e^{\theta X} \geq e^{\theta a}) \leq \underbrace{e^{-\theta a} E[e^{\theta X}]}$$

if $X_t \overset{d}{=} \text{Poisson}(\lambda t)$ then

$$= e^{\underbrace{-\theta a + t c(\theta)}_{(2)}}$$

To get best estimate, optimize wrt. θ.

$$(2) = -\theta a + t\lambda(e^{\theta} - 1)$$

take $\frac{d}{d\theta}(\cdots)$, set $= 0$;

"best" θ value.

when will it give good estimates?

If $a \gg \lambda t$ (e.g. $a = (\lambda + \varepsilon) t$

t is large).

estimates are $\leq e^{-\gamma(\varepsilon) t}$

$\gamma(\varepsilon) > 0$ if $\varepsilon > 0$.

☆ Exercises from Chapter 2

$\lambda = 1/2$

(2.1) $W \overset{d}{=} \exp(\text{rate} = 1/\text{mean} = 1/2)$

(a) $P(W > 2) = e^{-2\lambda} = e^{-1} = 1/e.$

(b) $P(W > 5 \mid W > 3) = P(W > 2) = 1/e.$

(2.4)

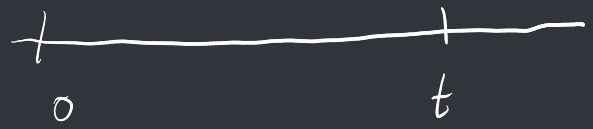
$\boxed{1}$ on at time 0

$\boxed{2}$ on @ t.

2 fails first if 1 not fail on $[0, t]$,

then 2 fails before 1.

W_i = time in use for i to fail.



$$P(W_1 > t) = e^{-\lambda_1 t}$$

given 1 working @ time t,

$$P(2 \text{ fails first}) = \lambda_2 / (\lambda_1 + \lambda_2)$$

$$P(2 \text{ fails before } 1) = e^{-\lambda_1 t} \cdot \lambda_2 / (\lambda_1 + \lambda_2)$$

(2.5) 3 fishing, each @ rate 2/hr.
How long till everyone has ≥1 fish?

N_t = # that haven't caught ≥1 fish yet

$N_0 = 3$

				expected time
N:	$3 \to 2$	@ rate	$2+2+2=6$	$1/6$
	$2 \to 1$	"	$2+2=4$	$1/4$
	$1 \to 0$	"	2	$1/2$

total expected time $= 1/6 + 1/4 + 1/2$
$= 11/12$

(2.8) $S, T \quad \exp(\lambda), \ \exp(\mu)$

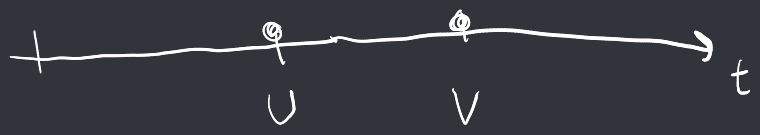
$U = \min(S,T) \overset{d}{=} \exp(\lambda+\mu)$

$V = \max(S,T)$

$W = V - U.$

w.p. $\lambda/_{\lambda+\mu}$, $U = S$

then $W \overset{d}{=} \exp(\mu)$



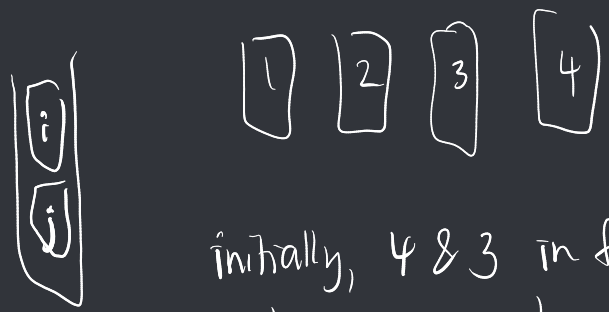
$\therefore \ \mu/_{\lambda+\mu}$ $U = T$

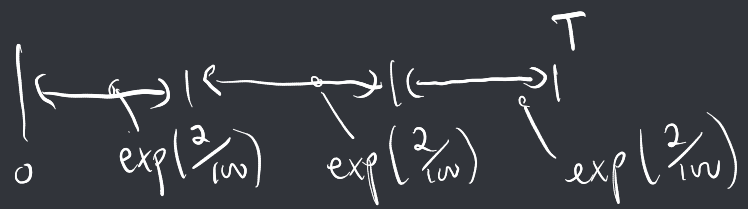
$W \overset{d}{=} \exp(\lambda)$

e.g. $\mathrm{Var}(W) = E[W^2] - E[W]^2$

& compute each from cond$^{\underline{tl}}$ dist$^{\underline{bn}}$.

2.12



initially, 4 & 3 in flashlight.
replace w/ 2 when one dies.
" " 1 " another " .

assume exponential
lifetime w/ mean = 100.